Link for the git repo containing the scripts:
https://github.com/Chetan-379/Geant4.git

Summary of the results can be found here:
Summary

Link for my webpage containing the output txt files:
https://cagrawal.web.cern.ch/folder/Geant4/

Link for my webpage containing the plots:
https://cagrawal.web.cern.ch/folder/Geant4/

Link for cleaned scripts of two photon studies till IMPETUS2 meeting:
https://github.com/Chetan-379/GEANT-4/tree/Cleaned_Till_IMPETUS2

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An Advanced tutorial on Geant4
https://videos.cern.ch/record/2737282

Some good slides for physics process in geant 4 (adv course):
https://indico.cern.ch/event/866056/contributions/3648977/attachments/1949289/3550230/ParticleProcesses.pdf

Some slides for cuts and physics process in geant4:
https://agenda.infn.it/event/22667/sessions/17787/attachments/71916/90481/Pandola_-_01_-_Physics_-_2.pdf

Git repo of Geant4 examples:
https://gitlab.cern.ch/geant4/geant4/-/tree/master/examples

Good Reference in Geant4 Manual for physics list:
https://geant4-userdoc.web.cern.ch/UsersGuides/PhysicsListGuide/html/physicslistguide.html

Youtube tutorial link:

https://youtube.com/playlist?list=PLLybgCU6QCGWgzNYOV0SKen9vqg4KXeVL&si=fFJK6UU2pDHyHHw3

Physics reference manual:
https://geant4-userdoc.web.cern.ch/UsersGuides/PhysicsReferenceManual/html/index.html

G4 user action slides:
https://indico.cern.ch/event/422168/contributions/1903302/subcontributions/170176/attachments/888132/1249408/userActions.ppt

An school (given by seema): https://indico.cern.ch/event/1376434/

Slides by Dr. Sunanda Banerjee:
https://indico.global/event/8005/contributions/72273/attachments/35511/66129/Simulation01.pdf

To understand Geant4 Architecture:
https://geant4-userdoc.web.cern.ch/UsersGuides/ForToolkitDeveloper/html/OOAnalysisDesign/Run/run.html

G4 Nist material list:
https://geant4-userdoc.web.cern.ch/UsersGuides/ForApplicationDeveloper/html/Appendix/materialNames.html

A good discussion on G4 regarding photon energy at different points inside a scintillator:
https://geant4-forum.web.cern.ch/t/how-precise-can-we-get-the-position-of-an-energy-deposition-deposited-due-to-a-specific-interaction/7728

Geant4 documentation for simulation of Photoelectric effect:
https://geant4.web.cern.ch/documentation/dev/prm_html/PhysicsReferenceManual/electromagnetic/gamma_incident/photoelectric/index.html

Seema's script for filling the entries of the tree into a histogram:
https://github.com/seemasharmafnal/HF_Analysis/tree/main/HFNtupleAnalysis

A good reference on G4:
https://indico.global/event/11040/contributions/103116/attachments/47526/90017/Pandola-Physics-1.pdf

A Good reference to read about cuts:
https://geant4-userdoc.web.cern.ch/UsersGuides/ForApplicationDeveloper/html/TrackingAndPhysics/thresholdVScut.html#prothres

Good slides for primary particle generation in geant4:
https://indico.cern.ch/event/294651/sessions/55918/attachments/552022/760639/PrimaryParticles.pdf

A Good reference to see what can be done with track and steps:
https://geant4.web.cern.ch/documentation/dev/faq_html/FAQ/tracksAndSteps.html

A Good discussion on the process defining an step:
https://geant4-forum.web.cern.ch/t/use-of-getprocessdefinedstep/7880

Documentation of G4 examples:
https://geant4-userdoc.web.cern.ch/Doxygen/examples_doc/html/

A useful slide on G4 examples:
https://indico.cern.ch/event/472305/contributions/1982314/attachments/1223730/1790337/Examples.pdf

Good slides for vis manager:
https://indico.cern.ch/event/776050/contributions/3241819/attachments/1789268/2914264/Visualization2-2.pdf

A good concise reference for interaction of photons with the material:
https://www.sciencedirect.com/science/article/abs/pii/B0122274105006347

G4Doc for multithreading (for toolkit developers):
https://geant4.web.cern.ch/documentation/pipelines/master/bftd_html/ForToolkitDeveloper/OOAnalysisDesign/Multithreading/mt.html

For writing the Geant4 Intro in thesis:
https://www.sciencedirect.com/science/article/pii/S0370157317301552

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References for Realistic measurement of Energy Deposits

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For Intrinsic time resolution: (By Seema on 15/3/25)

- https://arxiv.org/pdf/2106.04734
- https://arxiv.org/pdf/2401.00226

Concepts for optical photon interaction in the material:

- https://www.sciencedirect.com/science/article/pii/S0969806X23003079
- https://opengate.readthedocs.io/en/latest/generating_and_tracking_optical_photons.html

Efficiencies of different detector arrangements:
https://link.springer.com/article/10.1007/s41605-023-00399-9 (can be helpful to see which kind of plots to include in thesis)

Ray tracing in geant4:
https://www.sciencedirect.com/science/article/pii/S016890020801591X

SiPM-Based Scintillator Detector Simulation
https://arxiv.org/html/2403.02668v1

A good reference to look for geant 4 optical physics module:
https://indico.cern.ch/event/58317/contributions/2047457/attachments/992304/1411067/OpticalPhysics.pdf

A good discussion on defining surfaces:
https://geant4-forum.web.cern.ch/t/g4opticalsurface-correctly-setting-the-surface-properties/1143

Scintillation properties of PbWO4 Material: (**By Seema**)
https://www.sciencedirect.com/science/article/pii/S0168900202009166#:~:text=Compared%20to%20other%20promising%20fast,time%20%CF%84sc%3C60%20ns

A good report on Scintillator detectors: **(By Seema)**
https://www.tifr.res.in/~ehep2019/pdfs/exp05_scintillator.pdf

Geant4 example on Scintillators: **(by Seema)**
https://gitlab.cern.ch/geant4/geant4/-/tree/master/examples/advanced/composite_calorimeter

Reference for concise read of optical photons in geant4:
https://indico.esa.int/event/48/contributions/2528/attachments/2039/2387/G4SpaceForum_Optical.pdf

A good discussion on interaction of optical photons:
https://geant4-forum.web.cern.ch/t/issues-with-optical-properties-in-g4-v11-1-0/9854/6
A Reference to read about scintillators:
https://indico.ific.uv.es/event/7324/contributions/22257/attachments/11555/16268/Emssion_Tomography_Miles_N_and_John_N_Ch13.pdf

A good reference for scintillation photons:
https://indico.cern.ch/event/1019834/contributions/4280329/attachments/2210479/3956320/AH_OpticalPhotons_G4advanced_v2021_final.pdf

Slides for sensitive detector and touchable related stuff:
https://www.ge.infn.it/geant4/training/portland/readout.pdf

Geant4 Documentation for Boundary processes and Surface:

https://geant4-userdoc.web.cern.ch/UsersGuides/ForApplicationDeveloper/html/TrackingAndPhysics/physicsProcess.html#boundary-process

important for optical photons and surfaces etc.
https://indico.cern.ch/event/866056/contributions/3648975/attachments/1949282/3552948/AH_OpticalPhotons_G4advanced.pdf

To get Refractive index of a material:
https://refractiveindex.info/?shelf=main&book=Bi4Ge3O12&page=Williams

Taking the refractive index of PbWO4 from this paper:
https://www.researchgate.net/publication/2331895_Optical_properties_of_lead_tungstate_PbWO4_crystal_for_LHC_em-calorimetry (PWO N30, n_{ord})

papers based doing studies for scintillators:
https://www.mdpi.com/2227-9040/9/9/239
https://www.sciencedirect.com/science/article/pii/S0969804315000615

Peculiarities in the Simulation of Optical Physics with Geant4:
https://arxiv.org/pdf/1612.05162

Seems to be a good paper to browse through for transportation of optical photons:
https://arxiv.org/pdf/1802.09202

**********IMPORTANT**********

- **Excel Sheet containing Scintillation data: Scint_data**
- **Iterative fitting script: https://github.com/Chetan-379/GEANT-4/blob/main/test/Analyze/FitFunc.C#L204-L272**
- **My webpage containing the plots for studying the scintillator light response/resolution: https://cagrawal.web.cern.ch/folder/Geant4/resplots/**

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- For physical parameters of different materials (by Seema on 10/5/25):
  - PbWO4: https://www.sciencedirect.com/science/article/pii/S0022231321002465#:~:text=The%20absorption%20spectrum%20of%20the,H11%2F2%2C%20respectively.

    https://cds.cern.ch/record/260285/files/P00021913.pdf

https://cds.cern.ch/record/274714/files/SCAN-9501075.pdf

- Plastic (JPET): https://arxiv.org/pdf/2407.19465#:~:text=Emission%20spectra%20of%20the%20investigated,the%20spectra%20are%20the%20same.

- LYSO: https://iopscience.iop.org/article/10.1088/1742-6596/293/1/012004/pdf

- BGO: https://luxiumsolutions.com/sites/default/files/2021-09/BGO-material-data-sheet.pdf#:~:text=The%20material%20does%20not%20show,BGO%20is%20situated%20at%20480nm.

  https://cds.cern.ch/record/2772243/files/Fulltext%20from%20publisher.pdf

Useful Geant4 Material properties table reference:
https://apc.u-paris.fr/~franco/g4doxy4.11/html/classG4MaterialPropertiesTable.html

https://www.epic-crystal.com/scintillation-crystals/pbwo4-crystal.html

Geant4 documentation for material properties table:
https://geant4.web.cern.ch/documentation/pipelines/master/bfad_html/ForApplicationDevelopers/TrackingAndPhysics/physicsProcess.html#defining-material-properties

GAGG material properties:

Paper from palit et al: https://www.sciencedirect.com/science/article/pii/S0168900222012207
Efficiency analysis of photopeak:
https://www.sciencedirect.com/science/article/pii/S0168900224010556

A good reference to read about overview of Scintillator for practical purposes:
https://www.sciencedirect.com/science/article/pii/S0146641024000565

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For Detector geometry construction using GDML:

G4Example for reference: extended/persistency/gdml

GDML documentation: https://gdml.web.cern.ch/GDML/
https://gdml.web.cern.ch/GDML/doc/GDMLmanual.pdf

Slides from a Adv G4 course which might be useful:
https://indico.cern.ch/event/1304769/contributions/5487616/attachments/2679532/4757015/geometry-advanced-2023.pdf

An useful Adv. G4 course 2023: https://indico.cern.ch/event/1304769/timetable/#20231017

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For Polarisation in Geant4:

Some useful slides:
https://agenda.linearcollider.org/event/1265/contributions/1588/attachments/1035/1738/schaelicke_geant4.pdf

G4 documentation for polarisation:
https://geant4.web.cern.ch/documentation/dev/prm_html/PhysicsReferenceManual/electromagnetic/polarized_incident/index.html

G4 example for polarisation:
https://gitlab.cern.ch/geant4/geant4/tree/master/examples/extended/polarisation

Photon polarisation studies in JPET
JPET_polarisation.pdf

Seems to be a good reference for polarisation in compton scattering and other processes:
https://journals.aps.org/rmp/pdf/10.1103/RevModPhys.33.8

For gamma ray polarimeter, might be useful for studying polarisation from compton scattering:
https://journals.aps.org/pr/pdf/10.1103/PhysRev.78.551

Seems to be a good read for polarimetry using compton scattering:
https://www.nature.com/articles/s41598-025-87095-4

Seems to be a good reference for polarised compton:
https://www.sciencedirect.com/science/article/pii/0969806X9500472A

A reference for angular correlation in compton scattering (**by Seema**):
https://onedrive.live.com/?ls=true&cid=94D6F5343555954D&id=94D6F5343555954D%21s9897

6f8b57a8462caa6fcecb123f253e&parId=94D6F5343555954D%21s1bc0f1f1001e439fb1f3a12cba01c4ff&o=OneUp

Something relevant to compare our results on eta:
https://geant4-forum.web.cern.ch/t/how-to-code-geant4-to-generate-left-and-right-circulary-polarized-photons/3573/8

Another one with similar studies:
https://indico.global/event/8935/contributions/85640/attachments/39880/74184/iworld_poster_sato_revised.pdf

Non Maximal Entanglement:
https://koza.if.uj.edu.pl/files/5a5832ed6153b577391e37a1986aaa48/sciadv.ads3046.pdf

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References Related to positronium:

Paper on 3 photon Annihilation of electron and positron by A. Ore., JL Powell
https://journals.aps.org/pr/pdf/10.1103/PhysRev.75.1696

Physics list to simulate orthopositronium decay in G4:
https://geant4-userdoc.web.cern.ch/UsersGuides/AllGuides/html/PhysicsReferenceManual/decay/OrthoPositronium.html

QED and Fundamental symmetries of positronium:
https://www.actaphys.uj.edu.pl/fulltext?series=Reg&vol=50&page=1319

Atomic properties of positronium:
https://www.scopus.com/record/display.uri?eid=2-s2.0-85044600284&origin=inward&txGid=7e28dd356c99319e716778b2013a0465

Aspects of positronium physics:
https://inspirehep.net/files/772b66a0c7f467a1a7582e7f033dea71

Seems to be a good read: https://arxiv.org/abs/1807.04934 (Entanglement In Compton Scattering Processes)

Seems to be a good ref for review of oPs decay lifetime: Status of orthopositronium decay rate measurements - INSPIRE

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References related to JPET:

Geant4 Simulation of JPET:
https://github.com/JPETTomography/J-PET-geant4?tab=readme-ov-file

Good reference for polarisation measurement in PET in general:
https://www.sciencedirect.com/science/article/pii/S0168900222005514

Good Slides giving the overview of JPET:
https://indico.fis.agh.edu.pl/event/120/contributions/406/attachments/261/492/SharmaBEACH20220607_Final.pdf

A good reference giving the overview of JPET:
https://www.sciencedirect.com/science/article/pii/S016890022100437X

Photon polarisation studies in JPET
https://mail.google.com/mail/u/0/#sent?projector=1

Parity test in JPET:
https://koza.if.uj.edu.pl/files/4aca523c894b1afa184e85227c501b2d/s41467-021-25905-9.pdf

Test of single of plastic scintillator of JPET:
https://www.sciencedirect.com/science/article/pii/S0168900214009103

TOT method in JPET: TOT_JPET.pdf

Hit position reconstruction in JPET:LOR TOF reconstruction in JPET:
https://koza.if.uj.edu.pl/~marcin/files/a46ff0b49fad7347e92411a392cf2223/775.pdf

https://www.sciencedirect.com/science/article/pii/S0168900214008754

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ECAL test beam geometry of 5x5 crystals. https://gitlab.cern.ch/dejardin/h4sim

Regarding extracting deposited energy (By Alpana on 14/02/25):
I am sharing some examples how the energy deposited in the absorbers can be extracted. These can not be directly plugged into your setup but can give you an idea on how to go about it.
1. From standalone HGCAL simulation - this may work for you as it is CMSSW independent.

Looping over all elements in your volume and add the energy deposited in each block if it is an absorber
https://github.com/pfs/PFCal/blob/2922434ce888d40acdc832d13313c1e6a8835278/PFCalEE/src/SamplingSection.cc#L207

A volume is called absorber if it satisfies these conditions. These conditions can be updated as per your geometry.
https://github.com/pfs/PFCal/blob/master/PFCalEE/include/SamplingSection.hh#L112-L124

2. This is how we were doing in HGCAL TB simulation setup
https://gitlab.cern.ch/kalpana/hgcal_reconstruction_framework/-/blob/Gen_simstep/SimG4CMS/HGCalTestBeam/plugins/HGCPassive.cc#L435

3. This code I added to extract energy leaked out of the detector,
https://gitlab.cern.ch/kalpana/hgcal_reconstruction_framework/-/blob/Gen_simstep/SimG4CMS/HGCalTestBeam/plugins/HGCPassive.cc#L199-371

but you can modify it to get energy in the absorber energy if you can get access
https://gitlab.cern.ch/kalpana/hgcal_reconstruction_framework/-/blob/Gen_simstep/SimG4CMS/HGCalTestBeam/plugins/HGCPassive.cc#L317

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References related to KAPAE:

TIFR conference on quantum sensors:
https://indico.tifr.res.in/indico/conferenceTimeTable.py?confId=9404#all.detailed

Paper on Geant5: https://iopscience.iop.org/article/10.1088/1742-6596/396/2/022014/pdf

Imp commands to run geant4 on lxplus:
~~source /cvmfs/sft.cern.ch/lcg/views/LCG_105/x86_64-el9-gcc12-opt/setup.sh~~
source /cvmfs/sft.cern.ch/lcg/views/LCG_107/x86_64-el9-gcc14-opt/setup.sh